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Semester: III	Year : AY 2026-27
Subject: Programming with Java	
ASSIGNMENT No : 12	
TITLE : User-Defined Packages	

Problem Statement

Write a Java program to create and use user-defined packages.

Learning Objectives

To create and use user-defined packages for organizing Java programs.

Programme

Faculty:

```
package Experiment12.faculty;

public class Faculty {

    String name;
    String department;

    public Faculty(String name, String department) {
        this.name = name;
        this.department = department;
    }

    public void display() {
        System.out.println("----- Faculty Details-----");
        System.out.println("Faculty Name: " + name);
        System.out.println("Department: " + department);
    }
}
```

Student:

```
package Experiment12.student;
```

```

public class Student {

    String name;
    int rollNo;

    public Student(String name, int rollNo) {
        this.name = name;
        this.rollNo = rollNo;
    }

    public void display() {
        System.out.println("----- Student Details-----");
        System.out.println("Student Name: " + name);
        System.out.println("Roll No: " + rollNo);
    }
}

```

Library:

```

package Experiment12.library;

public class Book {

    int bookId;
    String title;
    String author;
    double price;

    public Book(int bookId, String title, String author, double price) {
        this.bookId = bookId;
        this.title = title;
        this.author = author;
        this.price = price;
    }

    public void display() {
        System.out.println("----- Book Details-----");
        System.out.println("Book ID: " + bookId);
        System.out.println("Title: " + title);
        System.out.println("Author: " + author);
        System.out.println("Price: " + price);
    }
}

```

1)

```
package Experiment12;
import Experiment12.faculty.Faculty;
import Experiment12.student.Student;

import java.util.Scanner;

public class Exp12a {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Student Name: ");
        String studentName = sc.nextLine();

        System.out.print("Enter Student Roll No: ");
        int rollNo = sc.nextInt();
        sc.nextLine();

        System.out.print("Enter Faculty Name: ");
        String facultyName = sc.nextLine();

        System.out.print("Enter Faculty Department: ");
        String department = sc.nextLine();

        Student s = new Student(studentName, rollNo);
        Faculty f = new Faculty(facultyName, department);

        System.out.println();

        s.display();

        System.out.println();

        f.display();

        sc.close();
    }
}
```

Output

```
PROBLEMS 35 OUTPUT DEBUG CONSOLE TERMINAL PORTS
in' 'Experiment12.Exp12a'
Enter Student Name: jenish
Enter Student Roll No: 11
Enter Faculty Name: pratik
Enter Faculty Department: cs

----- Student Details -----
Student Name: jenish
Roll No: 11

----- Faculty Details -----
Faculty Name: pratik
Department: cs
PS C:\Users\Admin\Desktop\Java> ^C
PS C:\Users\Admin\Desktop\Java>
```

2)

```
package Experiment12;

import java.util.Scanner;
import Experiment12.library.Book;

public class Exp12b {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Book ID: ");
        int bookId = sc.nextInt();
        sc.nextLine();

        System.out.print("Enter Book Title: ");
        String title = sc.nextLine();

        System.out.print("Enter Author Name: ");
        String author = sc.nextLine();

        System.out.print("Enter Book Price: ");
        double price = sc.nextDouble();

        Book b = new Book(bookId, title, author, price);

        System.out.println();

        b.display();

        sc.close();
    }
}
```

OUTPUT:

```
PS C:\Users\Admin\Desktop\Java> & 'C:\Program Files\Eclipse Adoptium\jdk-17.0.19-hotspot\bin\java.exe' '-agentlib:jdwp=transport=dt_socket,server=n,suspend=y,address=localhost:50627' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\Admin\AppData\Roaming\Code\User\workspaceStorage\66c35960ff10266939b1b6571e72db4a\redhat.java\jdt_ws\Java_9b8abd94\bin' 'Experiment12.Exp12b'
Enter Book ID: 1
Enter Book Title: jokeer
Enter Author Name: pratik'
Enter Book Price: 800

----- Book Details -----
Book ID: 1
Title: jokeer
Author: pratik'
Price: 800.0
PS C:\Users\Admin\Desktop\Java> |
```

Conclusion:

Both programs demonstrate the use of exception handling in Java to handle runtime errors effectively. By using try, catch, and appropriate exception handling techniques, the programs prevent abnormal termination and make the applications more reliable and user-friendly.